Fusion, Reranking and Metrics

1 Combining retrievers.

Reciprocal rank fusion is a method that combines several ranked lists by summing one over a constant plus the rank of each passage. Hybrid retrieval combines dense retrieval and BM25 to cover both meaning and exact terms. A cross-encoder is a model that reads the query and a passage together and scores their relevance. Reranking reorders the top passages with a slower and more accurate model.

2 Measuring retrieval.

Recall at k is the share of the relevant passages that appear in the top k results. Precision at k is the share of the top k results that are relevant. F1@k: The harmonic mean of precision and recall RankNet is a pairwise loss that measures the correctness of relative passage orders Reranking can improve precision at k but it cannot recover a passage the first stage never retrieved.

3 Combining in practice.

Reciprocal rank fusion needs no training and no weights, which makes it a safe default for hybrid retrieval. A cross-encoder is slower than a bi-encoder because it must read the query again for every passage. Reranking with a cross-encoder is applied only to the top passages of the fused list. Hybrid retrieval with reciprocal rank fusion usually beats BM25 alone and dense retrieval alone.

4 Reading the metrics.

Recall at k falls when the first stage misses relevant passages, and reranking cannot repair that. Precision at k rises when a cross-encoder moves relevant passages above irrelevant ones. F1@k balances recall at k and precision at k in one number.